NEWTON SCHOOL OF TECHNOLOGY
Data Visualization & Analytics — Capstone Project | 2024 - 2028

SECTOR: AI / TECHNOLOGY

WildChat Analytics Platform

Bussiness Intelligence Report

Behavioral Intelligence & Safety Monitoring for Conversational AI at Scale

PROBLEM STATEMENT
Behavioral Intelligence & Safety Analytics

DATASET
~8M Conversations (WildChat-1M)

PERIOD
April – May 2026

GROUP G-6 | SECTION A | Academic Year 2024 - 2028

SECTION 1 · **EXECUTIVE SUMMARY**

This report presents a comprehensive analytical investigation into the revenue dynamics and behavioral patterns of the WildChat conversational AI platform, drawing on approximately 8M validated point-of-sale interaction records spanning April 2023. The central research question — how do user engagement depth, model performance, geographic patterns, and safety signals interact to drive platform outcomes — is answered systematically through quantitative evidence derived from a rigorously engineered ETL pipeline.

Problem

Raw conversational telemetry lacked a unified operational intelligence framework. Without structured analytics, platform operators had no visibility into engagement drop-off patterns, language quality inequities, geographic safety risk concentration, or model-specific toxicity dynamics — all foundational inputs to product strategy and trust & safety governance.

Approach

A six-stage Python ETL pipeline transformed raw WildChat-1M data into ~8M production-grade conversation records. Nine analytical dimensions decomposed platform performance across engagement, quality, safety, geography, and temporal dynamics. All transformations were executed in Python (pandas, scikit-learn, NLTK, VADER) and visualised in an interactive three-module Tableau dashboard with 26 charts.

Key Findings

TOTAL CONVERSATIONS	USER ENGAGEMENT RATE	AVG TURN COUNT	DROP-OFF @ TURN 2
49,550	**53%**	**5.87**	**47%**
Validated records	*Conversations > 2 turns*	*Per conversation*	*Single-exchange sessions*

TOXICITY RATE	REPEAT USER RATE	AVG QUALITY SCORE	UNIQUE COUNTRIES
0.03%	**69%**	**7.94**	**148**
Avg composite score	*Conversation / IP ratio*	*Out of 16.2 max*	*Geographic coverage*

- China dominates volume with 12,128 conversations (24.5% of total) at only 1.10% toxicity, while Germany records 16.97% — 5.3× the platform average — despite ranking 4th by volume.
- GPT-4 delivers 31.7% higher quality scores than GPT-3.5 Turbo but attracts 63.7% more toxic interactions, indicating capability and safety risk move in the same direction.
- A power-user cohort of just 8% of unique IPs drives 59% of all conversation turns — extreme Pareto concentration creating platform fragility.
- Non-English speakers face a 0.5× higher unanswered rate; Chinese-language quality scores average 57% below English — a critical service equity gap across 35% of the user base.

Key Recommendations

- Deploy turn-two retention interventions (smart follow-up nudges) to reduce the 47% drop-off rate by 15–25%.
- Activate geo-targeted safety escalation for Germany, Brazil, and Latvia, which show disproportionately high toxicity-to-volume ratios.
- Invest in multilingual model quality to close the language equity gap affecting 17,000+ Chinese and Russian conversations.

Industry Landscape

The global conversational AI market exceeded USD 10 billion in 2023 and is projected to grow at a CAGR exceeding 23% through 2030. Platforms built on large language models now serve hundreds of millions of users across geographies, languages, and use-case verticals. This explosive growth has outpaced the development of operational governance frameworks, leaving a critical gap in how platforms monitor engagement quality, detect behavioral anomalies, and enforce content policy at scale.

Stakeholders & Decision-Makers

- Product Managers — need engagement KPIs, funnel visibility, and model performance benchmarks to prioritise feature development and capacity planning.
- Trust & Safety Officers — require real-time toxicity monitoring, anomaly detection, and geographic risk intelligence to enforce content policy and comply with emerging AI regulation.
- Research Scientists — need behavioral data on how different user populations use language models to inform training and alignment work.

Why This Problem Was Chosen

WildChat-1M represents one of the largest publicly available datasets of real-world LLM interactions with geographic and metadata enrichment. This project takes an operational intelligence lens — treating the data as a live production platform — to demonstrate how analytics disciplines generate business value from conversational AI telemetry.

Business Value

Quantifiable value includes: 15–25% projected improvement in engagement through turn-two interventions; 30% reduction in moderation overhead through automated geo-specific risk scoring; and potential 20% improvement in non-English user retention through language equity initiatives.

Formal Problem Definition

Conversational AI platforms operating at scale lack integrated visibility into the behavioral, linguistic, and safety dimensions of user interactions. Specifically, operators cannot simultaneously monitor (a) engagement depth and drop-off patterns, (b) differential service quality across languages and geographies, (c) toxicity trends and anomalous spikes, and (d) user segmentation and concentration risk — all from a unified analytical framework.

Project Scope

Dimension	In Scope	Out of Scope
Data	~50K conversations from WildChat-1M (April 2023 snapshot)	Real-time streaming, full 1M dataset
Analytics	Engagement, safety, quality, geography, segmentation, anomaly detection	Individual PII, competitive benchmarking
Output	Interactive Tableau dashboard, structured CSV exports	Deployed ML models, live API integration
Tooling	Python ETL, Tableau Desktop, Google Colab (GPU)	Custom front-end apps, cloud deployment

Success Criteria

- Clean dataset with zero critical missing values across primary dimensions
- KPI framework with at least five quantifiable, formula-based metrics linked to business decisions
- Three fully interactive Tableau dashboards with cross-filter and navigation capabilities
- Minimum eight data-backed insights expressed in decision language
- Three actionable recommendations with estimated business impact quantification

SECTION 4 · **DATA DESCRIPTION**

Primary Dataset: WildChat-1M (AllenAI, HuggingFace) — https://huggingface.co/datasets/allenai/WildChat-1M

Dataset	Rows	Key Fields	Primary Use
conversations_clean.csv	49,550	conversation_id, conv_turn_count, avg_sentiment, toxicity_score, model, country, language	Primary KPIs, EDA, segmentation
daily_kpis.csv	22	timestamp, conversation_count, avg_tokens, toxicity_rate	Time-series trend analysis, anomaly detection
geo_summary.csv	148	country, conversation_count, toxicity_rate	Geographic intelligence
messages_sample.csv	553	role, content, language, token_count, sentiment_score	Message-level NLP, quality analysis
prompt_categories.csv	10	prompt_category, count	Category distribution
radar_data.csv	6	Dimension, Actual, Target	Platform health radar chart

Key Field Definitions

Field	Type	Range / Values	Description
conv_turn_count	Integer	2–110 (mean: 5.87)	Message-reply pairs per conversation
avg_prompt_len	Decimal	0.8–3,296.5 (mean: 133)	Average token count of user messages
avg_sentiment	Decimal	-1.0 to +1.0 (mean: 0.169)	VADER compound sentiment score
toxicity_score	Decimal	0.0–0.579 (mean: 0.032)	Composite OpenAI moderation score
toxic_flag	Binary	0 or 1 (rate: 3.04%)	1 if toxicity_score > threshold
drop_off_flag	Binary	0 or 1 (rate: 47.24%)	1 if conversation ended at turn 2
response_quality_score	Decimal	0–16.2 (mean: 7.94)	Derived quality metric from model output
model	String	gpt-3.5-turbo-0301, gpt-4-0314	AI model used in conversation
language	String	68 unique values	Detected language of user messages
country	String	148 unique values	Geographic country from IP metadata

Data Limitations

- Geographic skew: China contributes 24.5% of conversations — VPN usage may distort true country-level figures.
- Language detection: 406 conversations (0.8%) classified as "Nolang" — excluded from language-specific analyses.
- Sample constraint: messages_sample.csv contains only 553 records, limiting message-level statistical power.
- Temporal window: 22 days of daily KPI data limits long-term trend extrapolation.

SECTION 5 · **DATA CLEANING & ETL PIPELINE**

All cleaning and transformation was executed in Python using pandas, numpy, scikit-learn, NLTK, and vaderSentiment. The pipeline is organised into six sequential stages executed via WildChat_Notebook.ipynb on Google Colab with GPU acceleration.

Stage	Objective	Records Affected	Outcome
1. Ingestion	Stream from HuggingFace Hub in 50K-row batches	20 batches × 50K rows	Memory-efficient full dataset access
2. Cleaning	Explosion, deduplication, type normalisation, geo validation	Duplicates removed; types unified	Zero duplicate IDs; clean schema
3. Text Preprocessing	Normalise content, detect language, count tokens, VADER sentiment	All 49,550 conversations	Sentiment scores + language labels applied
4. Feature Engineering	Compute toxicity composite, drop-off flag, quality score, user metrics	All records	12 derived analytical features created
5. Analytics & ML	K-means prompt clustering (k=5), user segmentation, geo clustering, daily KPIs	Clustered + segmented	5 prompt categories; 3 user segments
6. Exports	Generate Tableau-ready CSVs, summaries, dictionaries	6 output files	Production-ready analytical datasets

Key Assumptions

- Turn-2 drop-off treated as engagement failure proxy — some may represent successfully answered queries.
- Toxicity score reflects OpenAI moderation calibration, not universal content standards.
- Geographic attribution based on IP geolocation — VPN usage may introduce inaccuracies.

SECTION 6 · **KPI & METRIC FRAMEWORK**

Fourteen strategic KPIs were defined across five analytical dimensions. Each is directly traceable to a project objective and computed from verified cleaned data.

KPI	Formula	Observed Value	Business Decision Supported
User Engagement Rate	% of conversations with conv_turn_count > 2	53.0%	Feature prioritisation, onboarding UX investment
Average Turn Count	AVG(conv_turn_count) across all conversations	5.87 turns	Model capability assessment, session depth benchmarking
Toxicity Rate	AVG(toxicity_score) × 100	3.19%	Content moderation resource allocation
Drop-Off Rate	% of conversations with drop_off_flag = 1	47.24%	Retention strategy, prompt engineering
Repeat User Rate	COUNT(conversations) / COUNTD(hashed_ip)	~4.23× ratio	Platform stickiness, user loyalty assessment
Response Quality Score	AVG(response_quality_score)	7.94 / 16.2	Model selection ROI, quality benchmarking
Power User Concentration	Top 8% IPs as % of all turns	59% of turns	Dependency risk, loyalty programme justification
Language Gap	Non-English unanswered rate / English rate	0.5× ratio	Service equity, multilingual investment priority

7.1 Platform Summary KPIs

UNIQUE CONVERSATIONS	UNIQUE IPS	LANGUAGES	COUNTRIES	MODELS
49,550	**11,706**	**68**	**148**	**2**
Validated records	*Distinct user sessions*	*Detected codes*	*Geographic coverage*	*GPT-3.5 & GPT-4*

7.2 Geographic Distribution

Rank	Country	Conversations	Volume Share	Toxicity Rate	Risk Profile
1	China	12,128	24.5%	1.10%	High Volume · Low Toxicity
2	United States	5,606	11.3%	3.30%	High Volume · Moderate Toxicity
3	Russia	5,559	11.2%	3.27%	High Volume · Moderate Toxicity
4	Germany	2,239	4.5%	16.97%	Moderate Volume · CRITICAL Toxicity
5	France	1,499	3.0%	3.80%	Moderate Volume · Moderate Toxicity
6	Canada	1,397	2.8%	3.01%	Low-Moderate Toxicity
7	Hong Kong	1,396	2.8%	1.43%	Moderate Volume · Low Toxicity
8	Brazil	1,260	2.5%	3.33%	Moderate Volume · Moderate Toxicity
9	Morocco	1,177	2.4%	1.02%	Moderate Volume · Low Toxicity
10	India	1,153	2.3%	2.17%	Moderate Volume · Low-Moderate

Business Insight: Germany records 16.97% toxicity — 5.3× the platform average. China generates the highest volume at only 1.10% toxicity, demonstrating that geographic risk is entirely decoupled from geographic volume. Safety investment must follow risk concentration, not user headcount.

7.3 Model Version Performance

Metric	GPT-3.5-Turbo-0301	GPT-4-0314	Delta
Conversation Count	32,691 (66.0%)	16,859 (34.0%)	GPT-3.5 2× more common
Mean Quality Score	7.17	9.44	+31.7% for GPT-4
Mean Toxicity Score	0.026	0.043	+63.7% higher for GPT-4
Mean Turn Count	6.88 turns	3.91 turns	GPT-3.5 76% deeper sessions
Max Quality Score	14.80	16.18	GPT-4 higher ceiling

Business Insight: GPT-4 delivers 31.7% higher quality but attracts 63.7% more toxicity. More capable models draw more adversarial probing — model upgrades must be paired with upgraded safety calibration.

7.4 Prompt Category Distribution

Category	Conversations	Share (%)	Avg Sentiment
Casual	18,568	34.8%	Low positive (0.066)
Factual	12,026	23.1%	Moderate positive (0.352)
Other	10,752	16.3%	Mixed (0.208)
Coding	4,217	12.2%	Moderate positive (0.360)
Creative	3,987	12.0%	High positive (0.425)

Business Insight: Creative writing generates the highest positive sentiment (0.425) despite representing only 12% of interactions. Features surfacing creative use cases in onboarding flows could raise platform-wide satisfaction meaningfully.

7.5 Temporal Trends (22-Day Window)

Date	Conversations	Avg Tokens	Toxicity Rate	Notable
Apr 9, 2023	1,796	128.0	3.41%	Highest avg tokens — verbose sessions
Apr 15, 2023	1,570	106.3	6.53%	⚠ ANOMALY — Toxicity spike (Z > 2.6)
Apr 21, 2023	2,601	93.9	5.42%	⚠ ANOMALY — Secondary spike (Z ~ 1.8)
Apr 26, 2023	2,965	93.0	3.21%	Peak volume day of the window
Apr 30, 2023	1,668	112.4	3.18%	End-of-month volume decline

Business Insight: Two toxicity spikes detected — April 15 (6.53%, Z > 2.6) and April 21 (5.42%). Both coincide with lower-than-average volumes, suggesting concentrated adversarial sessions. Off-peak moderation staffing is critical.

8.1 User Segmentation

Segment	Definition	Share of IPs	Share of Turns	Avg Toxicity
Power Users	Top 10% by total turn count (≥50 turns)	~8%	~59%	2.80%
Regular Users	40th–90th percentile by turn count	~40%	~31%	3.20%
Casual Users	Bottom 50% by turn count (1–2 conversations)	~52%	~10%	3.40%

Business Insight: Power users (8%) generate 59% of all engagement. Loss of even a small fraction would dramatically reduce overall KPIs. Power-user loyalty architecture is operationally justified.

8.2 Anomaly Detection — Toxicity Z-Score

- April 15, 2023 (Z ≈ +2.6): Toxicity spike to 6.53% — the dataset's highest daily rate. Minimum volume day (1,570 conversations) suggests concentrated adversarial sessions.
- April 21, 2023 (Z ≈ +1.8): Secondary spike to 5.42% following a period of stable low toxicity.

8.3 Prompt Length vs Quality Correlation

Scatter analysis of avg_prompt_len vs response_quality_score reveals a logarithmic positive relationship. Prompts in the 300–1,500 token range consistently achieve quality scores above 13/16, while very short prompts (< 50 tokens) cluster below 8/16. The relationship plateaus beyond 1,500 tokens.

Business Insight: There is an optimal prompt-length zone (300–1,500 tokens). Platform-level prompt guidance tools could systematically shift users into this quality-optimal range.

8.4 Platform Health Radar Assessment

Dimension	Actual Score	Target Score	Gap	Priority
Engagement	68	80	−12 points	High
Drop-off (Inverted)	38	20	−18 points	Critical
Quality	72	85	−13 points	High
Retention	58	75	−17 points	Critical
Safety	85	90	−5 points	Low
Multilingual	42	60	−18 points	Critical

Safety is closest to target (85/90), while Drop-off, Retention, and Multilingual each carry 17–18 point gaps — defining the primary investment roadmap.

Dashboard Objective

The WildChat Analytics Tableau dashboard transforms processed data into an interactive decision-support environment for non-technical stakeholders, providing rapid access to platform KPIs, operational intelligence, and safety signals without requiring database access or coding skills.

HTML Prototype vs Final Tableau Implementation

Dimension	HTML Prototype (Initial)	Tableau Dashboard (Final)	Improvement
Data Connection	Hardcoded / static JSON	Live CSV connections (6 sources)	Real data integration
Interactivity	Basic tab switching	Cross-filter actions, parameters, tooltips	Full interaction
Visualisations	~8 basic charts	26 charts across 3 dashboards	3× analytical depth
Safety Analysis	Single toxicity rate	Z-score anomaly, heatmaps, rolling averages	Operational intelligence
Geographic View	Bar chart only	Country-level heatmap + toxicity overlay	Multi-dimensional geo
User Segmentation	Not present	K-means clusters, Pareto, retention curves	Advanced analytics

Three-Module Structure

Module 1 — Executive Summary (2 Pages)

Five KPI cards (53% Engagement, 5.9 Avg Turn Count, 0.03% Toxicity, 47% Drop-Off, 69% Repeat Users). Dual-axis Weekly Engagement & Drop-off Trend. Prompt Category Mix donut. Top Geographies table with toxicity colour-coding. Model Version Performance grouped bar. Platform Health Radar.

Module 2 — Operational Intelligence (2 Pages)

Four operational KPIs (47% Drop-off, 8% Power User Share, +35% Long Session Uplift, 0.5× Language Gap). Prompt Category bar. Response Quality by Language grouped bar. Session Duration histogram. Prompt Length vs Quality scatter. Activity Heatmap. Quality by Language box plots.

Module 3 — Safety & Trust (2 Pages)

Three safety KPIs (Toxicity 0.03%, Nocturnal Risk Window 22–02h, Adversarial Cluster 0.79). Toxicity Rate + Z-Score Anomaly Overlay dual-axis chart. Toxicity Concentration Heatmap (country × language). Sentiment Distribution diverging bar. Policy Violation trend chart.

SECTION 10 · INSIGHTS SUMMARY

The following ten insights are expressed in decision language — each tells a stakeholder what to think, believe, or do based on the evidence.

I1 Half of All Sessions Never Deepen

47.24% of conversations end after exactly two turns. This is a systematic failure to convert first-contact users into engaged sessions. Conversations with 8+ turns show 35% higher quality scores; the case for reducing turn-2 drop-off is quantifiable and compelling.

I2 Eight Percent of Users Drive Sixty Percent of Engagement

The top 8% of unique IPs generate 59% of all turns. This extreme concentration creates a fragile engagement model. Loss of a small fraction of power users would have outsized negative impact on all platform KPIs.

I3 Germany's Toxicity Rate Is an Outlier Demanding Dedicated Action

Germany records 16.97% toxicity — 5.3× the platform average — with 2,239 conversations. Volume-proportional moderation is insufficient. Germany, Brazil, and Latvia require dedicated geo-targeted safety policies.

I4 GPT-4 Is More Capable and More Dangerous

GPT-4 delivers 31.7% higher quality scores but attracts 63.7% higher toxicity scores. More powerful models draw more adversarial probing. Model upgrades must be accompanied by upgraded safety calibration.

I5 Non-English Users Receive Systematically Inferior Experiences

Non-English users face 0.5× higher unanswered rates. Chinese-language quality scores average 57% below English. Chinese and Russian speakers constitute 35% of the user base — this is a core equity and market access failure.

I6 Creative Engagement Generates the Highest User Satisfaction

Creative prompts generate the highest average positive sentiment (0.425) despite representing only 12% of interactions. Features surfacing creative use cases could meaningfully raise platform-wide satisfaction.

I7 Toxicity Spikes Are Predictably Concentrated in Off-Peak Hours

A nocturnal risk window between 22:00 and 02:00 shows a toxicity rate 2.8× the daytime average. Two of three Z-score anomaly events occurred during this window. Automated moderation escalation in these hours is high-impact and low-cost.

I8 Prompt Length Has a Quality Sweet Spot

Prompts in the 300–1,500 token range consistently achieve quality scores above 13/16. Very short prompts cluster below 8/16. Platform-level prompt guidance tools could systematically shift users into the quality-optimal zone.

I9 Platform Health Gaps Are Largest in Multilingual and Retention

Multilingual (42 vs. target 60, −18 pts) and Retention (58 vs. target 75, −17 pts) are furthest from goal. Safety is closest to target (85/90). This ranking should directly inform resource allocation decisions.

I10 China's Scale Masks Its Low Risk Profile

China generates 24.5% of all conversations at only 1.10% toxicity — the most scalable growth opportunity if language quality gaps are addressed. The contrast with Germany (15× higher toxicity, lower volume) confirms risk is not proportional to volume.

SECTION 11 · **RECOMMENDATIONS**

R1 **Priority: HIGH** Based on: I1, I2

Deploy Turn-Two Retention Interventions

Insight: 47.24% of conversations drop off after turn 2. Each 1% reduction represents approximately 10,000 additional engaged turns in a production environment at the platform's current scale.

Action: Implement smart follow-up prompt suggestions after the first AI response — context-aware nudges generated by a lightweight classifier identifying conversation intent, suggesting 2–3 natural continuation questions.

Implementation Feasibility: *Low — Software feature addition; no infrastructure change required.*

R2 **Priority: HIGH** Based on: I3

Activate Geo-Targeted Safety Escalation

Insight: Germany (16.97%), Brazil (3.33%), and Latvia show disproportionate toxicity rates versus their volume. A single uniform moderation policy treats high-risk and low-risk geographies identically.

Action: Build a three-tier geographic risk system (High / Moderate / Standard). High-tier countries trigger enhanced queue routing, stricter thresholds, and prioritised human review during nocturnal risk windows.

Implementation Feasibility: *Medium — Requires moderation system configuration; one-time policy calibration effort.*

R3 **Priority: HIGH** Based on: I5

Launch a Multilingual Quality Initiative

Insight: Non-English speakers face 0.5× higher unanswered rates and quality scores up to 57% below English. Chinese and Russian users represent 35% of the observed user base.

Action: Commission language-specific model fine-tuning starting with Mandarin Chinese and Russian. Implement a "language parity score" as a tracked KPI. Explore multilingual RLHF data collection from native-speaking annotators for the top five non-English languages.

Implementation Feasibility: *High — Requires ML engineering and annotation resources; multi-sprint delivery.*

R4 **Priority: MEDIUM** Based on: I2

Implement Power-User Loyalty Architecture

Insight: The top 8% of users generate 59% of platform engagement. Concentration is extreme even by Pareto standards and represents significant platform fragility.

Action: Design a power-user recognition programme including priority rate limits, early model access, dedicated feedback channels, and opt-in research participation incentives.

Implementation Feasibility: *Low — Programme design and scheduling; no capital infrastructure required.*

R5 **Priority: MEDIUM** Based on: I4

Implement Model-Specific Safety Calibration

Insight: GPT-4 conversations attract 63.7% higher toxicity scores than GPT-3.5, despite being more capable. A single unified moderation policy is insufficient.

Action: Develop separate thresholds and review queues for GPT-4 versus GPT-3.5. GPT-4 warrants a lower toxic_flag threshold (e.g., 0.07) and higher proportion routed to human review. Monitor toxicity shifts as a standard deployment gate.

Implementation Feasibility: *Low-Medium — Moderation system configuration and threshold tuning; no new infrastructure.*

SECTION 12 · IMPACT ESTIMATION

The following estimates are based on observed data patterns, industry benchmarks for conversational AI platforms, and logical extrapolation.

Recommendation	Metric	Baseline	Projected Outcome	Estimated Value
R1 — Turn-2 Retention	Drop-off rate	47.24%	35–40%	163–287 additional multi-turn sessions/day
R2 — Geo Safety Escalation	Unreviewed toxic content in high-risk regions	~70% pass-through	~40% pass-through	30% reduction in moderation workload
R3 — Multilingual Quality	Language quality parity score	57% gap (EN vs ZH)	30% gap within 12 months	17,000+ conversations improved quality
R4 — Power-User Loyalty	Power-user retention rate	Baseline unknown	+10% retention target	Preserves ~6% of total engagement volume
R5 — Model Safety Calibration	GPT-4 false-negative toxicity	~37% of flagged	~20–27%	20–30% reduction in undetected harmful content

Data Constraints

- Temporal window: 22 days of daily KPI data is insufficient for seasonal trend analysis or long-term forecasting.
- Message sample size: messages_sample.csv contains only 553 records — conclusions are directional, not definitive.
- Geographic attribution accuracy: IP-based geolocation is imprecise; VPN usage may distort country-level counts.
- Single time period: The April 2023 snapshot may not represent current platform behavior.

Methodological Assumptions

- Turn-2 drop-off as dissatisfaction proxy — some turn-2 conversations may represent successfully answered queries.
- K-means category labels are emergent; different parameters may yield different category boundaries.
- Quality score is derived, not user-rated; its correlation with actual satisfaction is not empirically validated.

What Cannot Be Concluded

- Causal claims: GPT-4 / toxicity correlation cannot be interpreted as causation without controlled experimentation.
- Individual user identification: All analyses are at hashed-IP level; no conclusions about specific individuals.
- Real-time applicability: The pipeline processes static snapshots; streaming infrastructure required for live use.

Enhanced Machine Learning

- Predictive drop-off model: Binary classifier (gradient boosted tree) to predict turn-2 drop-off from first-message features for real-time retention interventions.
- Topic modeling expansion: Apply BERTopic or LDA for richer semantic categorisation — potentially identifying 20–30 nuanced intent categories.
- Toxicity forecasting: ARIMA or Prophet time-series model on daily toxicity rates to generate proactive safety alerts 24–48 hours in advance.

Real-Time Pipeline

- Stream processing: Migrate from batch CSV exports to Apache Kafka → Spark Streaming feeding a live Tableau or Grafana dashboard with sub-minute refresh.
- Alert integration: Connect anomaly detection to PagerDuty or Slack for automated safety escalation during detected toxicity spikes.

Additional Data Sources Required

- Transaction timestamps (hour/minute) for intra-day demand analysis
- Customer / user identifier for cohort analysis and loyalty programme design
- Multi-year historical data (2022–2025) for trend validation and forecasting
- Human-rated quality assessments by native speakers for top five non-English languages

SECTION 15 · **CONCLUSION**

This report has delivered a comprehensive, evidence-based answer to the research question: how do user engagement depth, model performance, geographic patterns, and safety signals interact to drive platform outcomes in a conversational AI environment?

The analysis demonstrates that platform performance is driven by three structural realities. First, engagement depth is the primary lever: the 47% turn-2 drop-off rate represents both the platform's most significant failure point and its most recoverable opportunity. Second, geographic risk is entirely decoupled from geographic volume: Germany's 16.97% toxicity versus China's 1.10% — despite China generating 5× the conversations — means safety investment must follow risk concentration, not user headcount. Third, language equity is the platform's most underfunded dimension: with Chinese and Russian speakers constituting 35% of the user base but receiving systematically inferior quality, the multilingual gap is both a service failure and an untapped retention opportunity.

Five prioritised recommendations — turn-two retention interventions, geo-targeted safety escalation, multilingual quality investment, power-user loyalty architecture, and model-specific safety calibration — collectively address the engagement growth and content safety dimensions with quantifiable expected outcomes for each.

Value Delivered: From raw conversational telemetry to five actionable business recommendations — achieved through rigorous data engineering, systematic analytics, and business-language interpretation of every finding.

Appendix A — Data Dictionary

conversations_clean.csv (49,550 rows)

Column	Type	Description	Range / Values
conversation_id	String	Unique SHA-256 hashed conversation identifier	Unique per row
conv_turn_count	Integer	Number of message-reply pairs	2–110
avg_prompt_len	Decimal	Average token count of user messages	0.8–3,296.5
avg_sentiment	Decimal	VADER compound sentiment score	-1.0 to +1.0
country	String	Country from IP geolocation metadata	148 unique values
language	String	Detected language code (langdetect)	68 unique values
model	String	AI model used in conversation	gpt-3.5-turbo-0301, gpt-4-0314
hashed_ip	String	SHA-256 hash of user IP address	11,706 unique values
toxicity_score	Decimal	Composite OpenAI moderation score	0.0–0.579
toxic_flag	Integer	1 if toxicity_score > 0.1, else 0	0 or 1
drop_off_flag	Integer	1 if conv_turn_count = 2, else 0	0 or 1
response_quality_score	Decimal	Derived quality metric	0–16.2
prompt_category	String	K-means cluster label	Casual, Factual, Coding, Creative, Other

Appendix B — Tableau Calculated Fields

- User Engagement Rate: AVG(IF [conv_turn_count] > 2 THEN 1 ELSE 0 END)
- Drop-off Rate: AVG(IF [drop_off_flag] = 1 THEN 1 ELSE 0 END)
- Toxicity Rate: AVG([toxicity_score])
- Z-Score Toxicity: ([toxicity_rate] - WINDOW_AVG([toxicity_rate])) / WINDOW_STDEV([toxicity_rate])
- Is Anomaly: IF [Z-Score Toxicity] > 2.0 THEN "Anomaly" ELSE "Normal" END
- Rolling 7-Day Toxicity: WINDOW_AVG([toxicity_rate], -6, 0)
- Quality Band: IF [response_quality_score] >= 9 THEN "High" ELSEIF >= 6 THEN "Medium" ELSE "Low" END

Appendix C — Pipeline Configuration

- dataset_name: allenai/WildChat-1M | batch_size: 50,000 | max_batches: 20
- kmeans_k: 5 (prompt categories) | tfidf_max_feat: 5,000 | power_user_pct: 0.90
- toxicity_flag_threshold: 0.10 | drop_off_turn_threshold: 2

SECTION 17 · CONTRIBUTION MATRIX

The table below documents each team member's contributions across all project phases. Claims are verifiable through GitHub commit history, pull request records, and submitted artifact logs.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Stat Analysis	Tableau	Report Writing	PPT & Viva
Shaik Tajuddin (Lead)	✓ Lead	✓ Support	✓ Support	✓ Support	✓ Lead	✓ Lead	✓ Lead
Gaurav Meena	✓ Support	○	○	○	✓ Support	✓ Support	✓ Support
Hemamdhar Nath	✓ Support	○	○	○	○	✓ Support	○
Sushant Guri	✓ Lead	○	○	○	○	✓ Support	○
Jivit Rana	✓ Support	✓ Lead	✓ Lead	✓ Support	○	✓ Support	○
Parthraj S. Bhati	✓ Support	✓ Support	✓ Lead	✓ Lead	○	✓ Support	○
Samarth Chaudary	✓ Support	✓ Support	○	○	○	✓ Support	✓ Lead

Legend: ✓Lead = Primary contributor | ✓Support = Supporting contributor | ○ = No direct involvement

Declaration: We confirm the above contribution details are accurate and verifiable through GitHub Insights, pull request history, and submitted project artifacts.

Team Lead Signature: Shaik Tajuddin Date: April 29 2026